

Two dimensional random variable

Definition: Given a random experiment with sample space S , consider two random variables X_1 & X_2 on S . Then a function $X: S \rightarrow \mathbb{R}^2$ defined by

$$X(c) = (X_1(c), X_2(c)) \quad \forall c \in S$$

is called a two dimensional random variable.

The range of X is

$$\Theta = \{(x_1, x_2) \mid X_1(c) = x_1, X_2(c) = x_2 \quad \forall c \in S\}.$$

Example * Let a coin be tossed 3 times and our interest is in ordered number pair (number of H's on first two tosses, no. of H's on all three tosses).

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$X_1(c) = \underline{\text{no}}$ of heads on first two tosses.

$X_2(c) = \underline{\text{no}}$ of heads on all three tosses.

$$\therefore X = (X_1, X_2).$$

$$\mathcal{S} = \{ (2,3), (2,2), (1,2), (1,1), (0,1), (0,0) \}$$

CDF of two dimensional r.v

Let $X = (X_1, X_2)$ be random vector.

The CDF of X is a function,

$F_{X_1, X_2} : \mathbb{R} \times \mathbb{R} \rightarrow [0,1]$ defined by

$$F_{X_1, X_2}(x_1, x_2) = P[\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}]$$

Notation: $F_{X_1, X_2}(x_1, x_2) = P[X_1 \leq x_1, X_2 \leq x_2]$

$$\begin{aligned} \# \quad F_{X_1}(x_1) &= P[X_1 \leq x_1] \\ &= P(\{c \in \mathcal{S} \mid X_1(c) \leq x_1\}) \end{aligned}$$

$$\begin{aligned}
 F_{x_1, x_2}(x_1, \infty) &= P[\{x_1 \leq x_1\} \cap \{x_2 \leq \infty\}] \\
 &= P(\{c \in \mathcal{S} \mid x_1(c) \leq x_1 \text{ and } x_2(c) \leq \infty\}) \\
 &= P(\{c \in \mathcal{S} \mid x_1(c) \leq x_1\}) = F_x(x_1).
 \end{aligned}$$

Question: In Example 1, find $F_{x_1, x_2}(0.5, 2)$.

$$\begin{aligned}
 F_{x_1, x_2}(0.5, 2) &= P[\{x_1 \leq 0.5\} \cap \{x_2 \leq 1\}] \\
 &= P[\{\tau\tau\tau, \tau\tau H\} \cap \{\tau\tau\tau, \tau\tau H, H\tau\tau, \tau H\tau\}] \\
 &= P(\{\tau\tau\tau, \tau\tau H\}) = \frac{2}{8} = \frac{1}{4}.
 \end{aligned}$$

Joint probability mass function:

Let (X, Y) be two dimensional discrete random variable with range $\mathcal{D} \subseteq \mathbb{R}^2$.
 The joint pmf is a function:
 $p_{x_1, x_2} : \mathcal{D} \rightarrow [0, 1]$ defined by

$$p_{x_1, x_2}(x_1, x_2) = P[X_1 = x_1, X_2 = x_2] \\ = P[\{X_1 = x_1\} \cap \{X_2 = x_2\}]$$

Q. In Example $\otimes$ find $p_{x_1, x_2}(1, 1)$

$$p_{x_1, x_2}(1, 1) = P[\{X_1 = 1\} \cap \{X_2 = 1\}]$$

$$= P[\{HTH, THH, HTT, THT\} \cap \{HTT, THT, TTH\}]$$

$$= P[\{HTT, THT\}] = 2/8 = 1/4.$$

Properties of j.pmf :

1. For any (x_1, x_2) , $0 \leq p_{x_1, x_2}(x_1, x_2) \leq 1$

2. $\sum \sum_{(x_i, x_j) \in \mathcal{A}} p(x_i, x_j) = 1.$

3. $P[(X_1, X_2) \in B] = \sum \sum_{(x_i, x_j) \in B} p_{x_1, x_2}(x_i, x_j)$

The joint pmf is represented by table:

For example *

$x_1 \backslash x_2$	0	1	2	3
0	$\frac{1}{8}$	$\frac{1}{8}$	0	0
1	0	$\frac{2}{8}$	$\frac{2}{8}$	0
2	0	0	$\frac{1}{8}$	$\frac{1}{8}$

$$p(x_1, x_2) = P[X_1 = 0, X_2 = 0] = P(\{\tau\tau\tau\}) = \frac{1}{8}$$

$$p(0, 1) = P(\{\tau\tau H\}) = \frac{1}{8}$$

$$p(0, 2) = P(\emptyset)$$

Question: From above table find $P(X_1 \geq 2, X_2 \geq 2)$

$$\begin{aligned} \text{Sol: } P(X_1 \geq 2, X_2 \geq 2) &= P(\{X_1 \geq 2\} \cap \{X_2 \geq 2\}) \\ &= p(2, 2) + p(2, 3) = \frac{1}{8} + \frac{1}{8} = \frac{2}{8} \end{aligned}$$

Joint probability density function.

Let (X, Y) be two dimensional r.v.
 (X, Y) is said to be continuous two dimensional r.v if for every subset

$$P[(X, Y) \in B] = \iint_B f(x, y) dx dy.$$

The function $f(x, y)$ is called joint pdf.

Ex: Assume pdf of a continuous r.v (X, Y) is

$$f(x, y) = \begin{cases} 4xy e^{-(x^2+y^2)} & x \geq 0, y \geq 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $P\left(X > \frac{\sqrt{2}}{2}, Y > \frac{\sqrt{2}}{2}\right)$.

$$\text{Sol: } P\left(X > \frac{\sqrt{2}}{2}, Y > \frac{\sqrt{2}}{2}\right) =$$

$$= \int_{\frac{1}{\sqrt{2}}}^{\infty} \int_{\frac{1}{\sqrt{2}}}^{\infty} 4xy e^{-(x^2+y^2)} dx dy$$

$$= \int_{\frac{1}{\sqrt{2}}}^{\infty} 2y e^{y^2} dy \times \int_{\frac{1}{\sqrt{2}}}^{\infty} 2x e^{-x^2} dx$$

$$\text{Let } u = x^2 \quad du = 2x dx$$

$$\text{Also } x = \frac{1}{\sqrt{2}} \quad u = \frac{1}{2}$$

$$x = \infty \quad u = \infty$$

$$\begin{aligned} \int_{\frac{1}{\sqrt{2}}}^{\infty} 2x e^{-x^2} dx &= \int_{\frac{1}{2}}^{\infty} e^{-u} du = \left[\frac{e^{-u}}{-1} \right]_{\frac{1}{2}}^{\infty} \\ &= \frac{0 - e^{-\frac{1}{2}}}{-1} \\ &= e^{-\frac{1}{2}} \end{aligned}$$

$$\therefore P\left(X > \frac{\sqrt{2}}{2}, Y > \frac{\sqrt{2}}{2}\right) = \left(e^{-\frac{1}{2}}\right)^2 = \underline{\underline{e^{-1}}}$$

Properties of joint pdf

$$\text{i) } f_{x,y}(x,y) \geq 0 \quad \text{ii) } \iint_{\mathcal{D}} f_{x,y}(x,y) dx dy = 1.$$

The Marginal probability functions.

Discrete Case : Let (X,Y) be discrete random variable, with joint pdf $p_{x,y}$.

$\therefore$ The marginal pmf of X & Y are respectively

$$p_x(1) = p_{x,y}(1,0) + p_{x,y}(1,1)$$

$$p_x(\omega) = \sum_{y < \infty} p_{x,y}(\omega, y) \quad \begin{matrix} + p_{x,y}(1,2) \\ + p_{x,y}(1,3) \end{matrix}$$

$$p_y(\omega) = \sum_{x < \infty} p_{x,y}(x, \omega)$$

Question: Consider Example 1, where
 $X := \underline{\text{no}}$ of heads in first 2 tosses
 $Y := \underline{\text{no}}$ of heads in all three tosses.
 Determine the marginal pdf of X and Y .

Sol: The joint pmf table is given by

$x \backslash y$	0	1	2	3	<u>p_x</u>
0	$\frac{1}{8}$	$\frac{1}{8}$	0	0	$p_x(0) = \frac{1}{4}$
1	0	$\frac{2}{8}$	$\frac{2}{8}$	0	$p_x(1) = \frac{1}{2}$
2	0	0	$\frac{1}{8}$	$\frac{1}{8}$	$p_x(2) =$
	$p_y(0)$ $= \frac{1}{8}$	$p_y(1)$ $= \frac{3}{8}$			

Continuous Case: Let $f(x,y)$ be the joint pdf of continuous two-dimensional r.v (X,Y) . The marginal pdf of X and Y are respectively,

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy \qquad h(y) = \int_{-\infty}^{\infty} f(x,y) dx.$$

Question: Suppose (X,Y) is a two-dimensional continuous r.v with joint pdf

$$f(x,y) = 2(x+y-2xy), \quad 0 \leq x \leq 1, 0 \leq y \leq 1. \\ = 0, \quad \text{else where}$$

Find marginal pdf of X & Y .

Sol: The marginal pdf of X is

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy$$

$$= \int_0^1 2(x+y-2xy) dy = 2 \left[xy + \frac{y^2}{2} - xy^2 \right]_0^1$$

$$= 2 \left[x + \frac{1}{2} - x \right] = 1.$$

The marginal pdf of Y is $h(y) = \int_{-\infty}^{\infty} f(x,y) dx = 1$

Independent random variables:

Let (X, Y) be 2-dimensional r.v.

X & Y are said to be independent if

- Discrete case: $p_{x,y}(x,y) = p_x(x) p_y(y) \quad \forall (x,y) \in \mathcal{D}$
- Continuous case: $f(x,y) = g(x) h(y) \quad \forall x,y \in \mathcal{D}$

Question: Let X & Y be life lengths of two electronic devices. Suppose their joint pdf is $f(x,y) = e^{-(x+y)} \quad x \geq 0, y \geq 0$. Show X & Y are independent.

$$\text{Sol: } g(x) = \int_0^{\infty} e^{-(x+y)} dy = e^{-x} \int_0^{\infty} e^{-y} dy$$

$$= -e^{-x} [e^{-y}]_0^{\infty}$$

$$= -e^{-x} [0 - 1] = e^{-x}$$

Similarly $h(x) = e^{-y}$.

$$\therefore g(x)h(x) = f(x,y)$$

Expectation of 2D RV :

Let (X, Y) be 2D RV, then expectation of X and Y are given by

Discrete case:

$$E(X) = \sum_{i=1}^{\infty} x_i p_x(x_i) = \sum_{i=1}^{\infty} x_i \sum_j p(x_i, y_j)$$

$$= \sum_i \sum_j x_i p(x_i, y_j)$$

$$E(Y) = \sum_{i=1}^{\infty} y_i p_y(y_i) = \sum_j \sum_i y_j p(x_i, y_j)$$

Continuous case:

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x g(x) dx = \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f(x, y) dy dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dy dx \end{aligned}$$

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y h(y) dy = \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \end{aligned}$$

Properties of Expectation

Let (X, Y) be two dimension random variable.

i) $E(aX + bY) = aE(X) + bE(Y)$, $a, b \in \mathbb{R}$

ii) If X & Y are independent

- $E(XY) = E(X)E(Y)$

- $V(X+Y) = V(X) + V(Y)$

Proof : ii) • $E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x,y) dx dy$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy g(x) h(y) dx dy \quad [X \text{ \& } Y \text{ are independent}]$$

$$= \int_{-\infty}^{\infty} x g(x) dx \times \int_{-\infty}^{\infty} y h(y) dy = E(X)E(Y)$$

$$\bullet V(X+Y) = E[(X+Y)^2] - [E(X+Y)]^2$$

$$= E(X^2 + Y^2 + 2XY) - [E(X) + E(Y)]^2$$

$$= E(X^2) + E(Y^2) + 2E(XY) - E(X)^2 - E(Y)^2 - 2E(X)E(Y)$$

$$= [E(X^2) - E(X)^2] + [E(Y^2) - E(Y)^2]$$

$$= V(X) + V(Y)$$

Correlation Coefficient :

Let X and Y be two random variables
Correlation coefficient of X & Y is

$$\rho_{xy} = \frac{E[(X - E(X))(Y - E(Y))]}{\sqrt{V(X) \cdot V(Y)}}$$

Numerator is called covariance of X & Y is denoted $\text{COV}(X, Y)$

- When $\rho_{xy} > 0$, if X increases, then Y also increases.
- When $\rho_{xy} < 0$, if X increases, then Y decrease.

In other words, ρ tells that deviation from respective means of X & Y either move together or opposite.

Result:
$$\rho_{xy} = \frac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}}$$

Result: If X and Y are independent, then $\rho_{xy} = 0$.

Remark: If $\rho_{xy} = 0$, then X & Y need not be independent.

Ex: Let X be r.v., with pdf

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } -1 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Consider $Y = X^2$ to be another r.v.

$$E(XY) = E(X^3) = \int_{-1}^1 x^3 f(x) dx = \int_{-1}^1 x^3 \times \frac{1}{2}$$

$$= \frac{1}{2} \left[\frac{x^4}{4} \right]_{-1}^1 = 0$$

$$E(X) = \int_{-1}^1 x f(x) dx = \int_{-1}^1 x \cdot \frac{1}{2} dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_{-1}^1 = 0$$

$$\therefore E(XY) = 0 = E(X) \cdot E(Y) \Rightarrow \rho_{XY} = 0$$

$\therefore X$ & Y are dependent.

Remark: If $\rho_{XY} = 0$, we say X & Y are uncorrelated.

Properties of ρ_{XY}

1. $-1 \leq \rho \leq 1$

2. If X & Y are linearly related, that is, if $Y = a + bX$ for some real numbers a and b , then $\rho = \pm 1$.

Problems:

1. The random vector (X, Y) has a joint pdf given by,

$$f(x, y) = x + y, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

Compute correlation b/w X & Y .

$$\text{Sol: } \rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}}$$

$$E(X) = \int_{-\infty}^{\infty} x g(x) dx, \text{ where } g(x)$$

is marginal pdf of X , given by

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 (x+y) dy$$

$$= \left[xy + \frac{y^2}{2} \right]_0^1 = x + \frac{1}{2}$$

$$\therefore E(X) = \int_{-\infty}^{\infty} x \left(x + \frac{1}{2} \right) dx = \int_0^1 \left(x^2 + \frac{x}{2} \right) dx$$

$$= \left[\frac{x^3}{3} + \frac{x^2}{4} \right]_0^1 = \frac{1}{3} + \frac{1}{4} = \underline{\underline{\frac{7}{12}}}$$

$$E(Y) = \int_{-\infty}^{\infty} y h(y) dy, \text{ where } h(y) \text{ is}$$

marginal pdf of Y given by,

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 (x+y) dx$$

$$= \left[\frac{x^2}{2} + xy \right]_0^1 = \frac{1}{2} + y$$

$$\begin{aligned} \therefore E(Y) &= \int_0^1 y \left(\frac{1}{2} + y \right) dy = \left[\frac{y^2}{4} + \frac{y^3}{3} \right]_0^1 \\ &= \underline{\underline{\frac{7}{12}}} \end{aligned}$$

$$E(XY) = \int_0^1 \int_0^1 xy (x+y) dx dy$$

$$= \int_0^1 y \int_0^1 [x^2 + xy] dx dy$$

$$= \int_0^1 y \left[\frac{x^3}{3} + \frac{x^2 y}{2} \right]_0^1 dy = \int_0^1 y \left(\frac{1}{3} + \frac{y}{2} \right) dy$$

$$= \left[\frac{y^2}{6} + \frac{y^3}{6} \right]_0^1 = \frac{2}{6} = \frac{1}{3}$$

$$V(X) = E(X^2) - E(X)^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 g(x) dx$$

$$= \int_0^1 x^2 \left(x + \frac{1}{2}\right) dx = \int_0^1 \left(x^3 + \frac{x^2}{2}\right) dx$$

$$= \left[\frac{x^4}{4} + \frac{x^3}{6} \right]_0^1 = \frac{1}{4} + \frac{1}{6} = \frac{10}{24} = \frac{5}{12}$$

$$\therefore V(X) = \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{60-49}{144} = \frac{11}{144}$$

$$E(Y^2) = \frac{11}{144}$$

$$\therefore \rho_{XY} = \frac{\frac{1}{3} - \frac{7}{12} \times \frac{7}{12}}{\frac{11}{144}} = \frac{\frac{48}{144} - \frac{49}{144}}{\frac{11}{144}}$$

$$= -\frac{1}{11}$$

2. Given $E(XY) = 43$, $P(X=x_i) = 1/5$ and $P(Y=y_j) = 1/5$. Find ρ .

X	1	3	4	6	8
Y	1	2	24	12	5

$$\text{Sol: } E(X) = \sum_{x_i \in \mathcal{X}} x_i p(x_i)$$

$$= \frac{1 + 3 + 4 + 6 + 8}{5} = \frac{22}{5}$$

$$E(Y) = \sum y p(y) = \frac{1 + 2 + 24 + 12 + 5}{5} = \frac{44}{5}$$

$$E(X^2) = \frac{1 + 9 + 16 + 36 + 64}{5} = \frac{126}{5}$$

$$E(Y^2) = \frac{1 + 4 + 576 + 144 + 25}{5} = \frac{750}{5} = 150$$

$$\therefore V(X) = E(X^2) - E(X)^2$$

$$= \frac{126}{5} - \left(\frac{22}{5}\right)^2 = \frac{630}{25} - \frac{484}{25} = \frac{146}{25}$$

$$V(Y) = E(Y^2) - E(Y)^2 = \frac{750}{5} - \left(\frac{44}{5}\right)^2$$

$$= \frac{3750 - 1936}{25} = \frac{1814}{25} = 72.56.$$

$$\rho = 43 - \frac{22}{5} \times \frac{44}{5}$$

$$\sqrt{\frac{146}{25} \times \frac{1814}{25}}$$

$$= \frac{1075 - 968}{25}$$

$$\frac{264844}{25}$$

$$= \frac{107}{514.63} = 0.20791$$

3. If x, y, z are uncorrelated r.v with standard deviation 5, 12 & 9 respectively. If $U = x + y$, $V = y + z$. Evaluate ρ b/w U and V .

Sol: Given $0 = \text{Cov}(x, y) = \text{Cov}(y, z) = \text{Cov}(x, z)$

$$\rho_{UV} = \frac{E(UV) - E(U)E(V)}{\sqrt{\text{Var}(U)} \sqrt{\text{Var}(V)}}$$

$$= \frac{E((X+Y)(Y+Z)) - E(X+Y)E(Y+Z)}{\sqrt{V(X+Y)} \sqrt{V(Y+Z)}}$$

$$= \frac{E(XY + XZ + Y^2 + YZ) - [E(X) + E(Y)][E(Y) + E(Z)]}{\sqrt{[V(X) + V(Y)][V(Y) + V(Z)]}}$$

$$= \frac{E(XY) + E(XZ) + E(Y^2) + E(YZ) - \{E(X)E(Y) + E(X)E(Z) + E(Y)^2 + E(Y)E(Z)\}}{\sqrt{(25 + 144)(144 + 81)}}$$

$$= \frac{E(Y^2) - E(Y)^2}{13 \times 15} = \frac{V(Y)}{195} = \underline{\underline{\frac{144}{195}}}$$

Question: Two independent r.v X_1 & X_2 has mean 5, 10 and variance 4 and 9 respectively. Find the covariance b/w $u = 3x_1 + 4x_2$ and $v = 3x_1 - x_2$.

Sol: Let $U = 3X_1 + 4X_2$ and
 $V = 3X_1 - X_2$ be r.v.

$$\begin{aligned} &= E(UV) - E(U)E(V) \\ &= E[(3X_1 + 4X_2)(3X_1 - X_2)] - E(3X_1 + 4X_2)E(3X_1 - X_2) \\ &= E(9X_1^2 - 3X_1X_2 + 12X_1X_2 - 4X_2^2) - \left[(3E(X_1) + 4E(X_2)) \right. \\ &\quad \left. (3E(X_1) - E(X_2)) \right] \\ &= 9E(X_1^2) + 9E(X_1X_2) - 4E(X_2^2) \\ &\quad - \{ 9E(X_1)^2 - 3E(X_1)E(X_2) + 12E(X_1)E(X_2) - 4E(X_2)^2 \} \\ &= 9E(X_1^2) + 9E(X_1X_2) - 4E(X_2^2) \\ &\quad - 9E(X_1)^2 + 3E(X_1)E(X_2) - 12E(X_1)E(X_2) + 4E(X_2)^2 \\ &= 9(E(X_1^2) - E(X_1)^2) - 4(E(X_2^2) - E(X_2)^2) \\ &\quad + 9(E(X_1X_2) - E(X_1)E(X_2)) \\ &= 9 \times 4 - 4 \times 9 = \underline{\underline{0}} \end{aligned}$$

Remark: U & V are uncorrelated but not independent.

Q. Prove that $V(aX+bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X, Y)$.

$$\begin{aligned}\text{So! } V(aX+bY) &= E[(aX+bY)^2] - [E(aX+bY)]^2 \\&= E[a^2 X^2 + b^2 Y^2 + 2abXY] - [aE(X) + bE(Y)]^2 \\&= a^2 E(X^2) + b^2 E(Y^2) + 2ab E(XY) \\&\quad - [a^2 E(X)^2 + b^2 E(Y)^2 + 2ab E(X)E(Y)] \\&= a^2 [E(X^2) - E(X)^2] + b^2 [E(Y^2) - E(Y)^2] \\&\quad + 2ab [E(XY) - E(X)E(Y)] \\&= a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X, Y)\end{aligned}$$